

FACE RECOGNITION Using YOLO

SOMNATH BERA AND SUKRITI SAHOO

This device integrates real-time face detection with a 5-point facial landmarking system to identify known and unknown faces. It serves multiple applications, from a face recognition-based doorbell to an attendance and time management system for MSMEs with fewer than 100 employees. Table 1 details the required components, while Fig. 1 shows the author's prototype.

MaixCam dev board

The MaixCam is popular for edge AI applications and prototyping new AI algorithms. It is well-supported with libraries like MaixPy, a Python-based framework for K210 boards, making it accessible to both beginners and experts.

It uses GPIO pins to control external devices, such as a gate mechanism for employee access.

BILL OF MATERIALS

Components	Quantity
MaixCam development board	1
Red and green LED	2
5V, 2A adaptor for power	1

LEDs are used as visual indicators. The GPIO pin map for the MaixCam board is used to control peripherals like LEDs or servo motors for the gate mechanism. A green LED lights up for recognised (known) faces, while a red LED lights up for unrecognised (unknown) faces.

Five-point face recognition system

A five-point face recognition system is used here. The system focuses on identifying five critical landmarks on a face. These five points are the two eyes, the nose tip, and the two corners of the mouth. These landmarks are essential for alignment and recognition in most face detection algorithms.

YOLOv8 is the latest version of the YOLO (You Only Look Once) family. It is faster, more accurate, and more efficient. Face detection locates faces within an image, while landmark detection pinpoints five key facial points. Recognition then matches the detected face with a stored database.

Operating with YOLOv8 offers

several advantages, including high accuracy, ensuring precise detection even in low-light or cluttered environments. It is highly efficient, running in real time on embedded systems. Additionally, its versatility allows it to detect objects, landmarks, and more within a single model.

Hardware

For AI processing, the Sipeed micro-controller is used, featuring the following specifications: a dual-core CPU with a Big Core running at 1GHz (RISC-V C906/ARM A53) and a small core at 700MHz (RISC-V C906/ARM A53). It includes an NPU delivering 1 TOPS@INT8 (1 TOPS = 1 trillion \ 10^{12} operations per second), 256MB DDR3 RAM, and a TFT touchscreen.

The system runs a Linux-based operating system specifically designed for Kendryte K210 boards. Key features include integrated support for YOLOv8, a headless mode for efficient remote operations, Python 3.11 for scripting and automation, and an SSH daemon for secure remote access. This setup is ideal for real-time face recognition,



Fig. 1: Authors' prototype